SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA5111
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

I Kala Gowri Shankar (192325109) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A
Analytical Problem Solving

2. Block Cipher Mode Evaluation

Given plaintext blocks:
P=[1010,1010,1111,1010], Key = K=1100

a. Encrypt using ECB and CBC modes (Assume IV = 0000) b.
Compare the ciphertext outputs
c. Analyze which mode hides patterns better and explain why

Problem Understanding

Given:

- Plaintext blocks:
  - P = **1010**
  - P = **1010**
  - P = **1111**
  - P = **1010**
- Key (K) = **1100**
- IV = **0000** (for CBC)

Assume encryption is performed using a simple XOR operation:

Ciphertext = Plaintext ⊕ Key

Method Selection

- **ECB (Electronic Codebook):**
  - Each plaintext block is encrypted independently.
  - Formula:

    C = P ⊕ K
- **CBC (Cipher Block Chaining):**
  - Each plaintext block is XORed with the previous ciphertext before encryption.
  - Formula:
    - $C = (P \oplus IV) \oplus K$
    - $C = (P \oplus C) \oplus K$

Solution Process

(a) ECB Encryption

Plaintext	Key	Ciphertext = P ⊕ K
1010	1100	0110
1010	1100	0110
1111	1100	0011
1010	1100	0110

ECB Ciphertext

[0110, 0110, 0011, 0110]

CBC Encryption

Block 1

$P \oplus IV$

1010 ⊕ 0000 = 1010

Encrypt

1010 ⊕ 1100 = **0110**

C = **0110**

Block 2

$P \oplus C$

1010 ⊕ 0110 = 1100

Encrypt

1100 ⊕ 1100 = **0000**

C = **0000**

Block 3

$P \oplus C$

1111 ⊕ 0000 = 1111

Encrypt

1111 ⊕ 1100 = **0011**

C = **0011**

Block 4
P ⊕ C
1010 ⊕ 0011 = 1001
Encrypt
1001 ⊕ 1100 = **0101**
C = **0101**

CBC Ciphertext
[0110, 0000, 0011, 0101]

Accuracy of Calculation
ECB
1010 ⊕ 1100 = 0110
1010 ⊕ 1100 = 0110
1111 ⊕ 1100 = 0011
1010 ⊕ 1100 = 0110
CBC
Block1:
(1010⊕0000)=1010
1010⊕1100=0110

Block2:
1010⊕0110=1100
1100⊕1100=0000

Block3:
1111⊕0000=1111
1111⊕1100=0011

Block4:
1010⊕0011=1001
1001⊕1100=0101
All XOR calculations are correct.

Interpretation of Results
(b) Comparison of Ciphertext Outputs

Mode	**Ciphertext**
ECB	[0110, 0110, 0011, 0110]
CBC	[0110, 0000, 0011, 0101]

(c) Pattern Analysis

- In **ECB**, identical plaintext blocks produce identical ciphertext blocks.
- Since **1010** appears three times, the ciphertext **0110** also appears three times.
- This reveals patterns in the encrypted data.

- In **CBC**, each block depends on the previous ciphertext.
- Even though **1010** is repeated, the ciphertext values become different.
- This hides patterns and improves security.

Conclusion:
CBC mode hides patterns better than ECB because each ciphertext block depends on both the current plaintext block and the previous ciphertext block.

3. DES vs 3-DES vs Blowfish Performance Study
Given:

a. DES key size = 56 bits, time = 2 ms/block
b. 3-DES key size = 168 bits, time = 6 ms/block
c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

i. Calculate total encryption time for each algorithm
ii. Analyze trade-offs between security and performance
iii. Recommend the best algorithm for secure real-time communication

Problem Understanding
Given:

Algorithm	Key Size	Time per Block
DES	56 bits	2 ms
3-DES	168 bits	6 ms
Blowfish	128 bits	1 ms

Number of blocks = **1000**
Need to:

1. Calculate total encryption time.
2. Compare security and performance.
3. Recommend the best algorithm for secure real-time communication.

Method Selection
Formula:
Total Encryption Time = Number of Blocks × Time per Block

Solution Process
DES
Time per block = 2 ms
Total Time
= 1000 × 2
= **2000 ms**
= **2 seconds**

3-DES
Time per block = 6 ms
Total Time
= 1000 × 6
= **6000 ms**

= **6 seconds**

Blowfish
Time per block = 1 ms
Total Time
= 1000 × 1
= **1000 ms**
= **1 second**

Accuracy of Calculation

Algorithm	**Formula**	**Total Time**
DES	1000 × 2	2000 ms (2 s)
3-DES	1000 × 6	6000 ms (6 s)
Blowfish	1000 × 1	1000 ms (1 s)

Calculations are correct.

Interpretation of Results
(ii) Security vs Performance

Algorithm	**Security**	**Performance**	**Remarks**
DES	Low	Medium	56-bit key is vulnerable to brute-force attacks.
3-DES	Very High	Slow	Strong security but three encryption rounds increase execution time.
Blowfish	High	Fast	Provides good security with the fastest encryption time among the three.

(iii) Recommendation
For **secure real-time communication**, **Blowfish** is the best choice because:

- Encrypts data in only **1 second** for 1000 blocks.
- Uses a **128-bit key**, offering stronger security than DES.
- Faster than both DES and 3-DES, making it suitable for applications requiring low latency.
- Provides a good balance between **security** and **performance**.

Final Recommendation: Blowfish is the preferred algorithm for secure real-time communication due to its high security and fast encryption speed.

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation